

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I Kamalli shree V (192525030) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of AI Application	Provide the name of the latest AI application selected and the corresponding AI domain.
2	Domain Selection	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	Introduction and Background	Describe the AI application and provide an overview of its purpose and importance.
4	Problem Identification	Explain the real-world problem or challenge addressed by this AI application.
5	Need for AI Solution	Analyze why AI is required and how it improves traditional approaches.
6	AI Technologies Used	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	Working Mechanism	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	Data Sources and Processing	Describe the type of data used and how the data is processed for intelligent decision-making.
9	Real-World Implementation	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	Impact and Benefits	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	Challenges and Limitations	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	Future Scope and Emerging Trends	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	Conclusion	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	References	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (59–50%)
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.

AI-Based Elderly Companion Robot with Intelligent Medication Reminder and Health Monitoring System

Abstract:

The increasing elderly population has created a growing demand for intelligent healthcare solutions that support independent living. Many senior citizens forget to take medicines, miss appointments, or face medical emergencies while living alone. This proposed AI-based companion robot provides personalized medication reminders, monitors vital signs, detects falls, and communicates with caregivers during emergencies.

Problem Statement:

- Elderly people often forget medications and appointments.
- Traditional reminder systems only generate alarms and cannot verify medication intake.
- Lack of continuous health monitoring increases emergency risks.
- Need for an intelligent, adaptive, and healthcare assistant.

Latest AI Application:

It represents a modern AI healthcare application by combining:

- Machine Learning
- Natural Language Processing (NLP)
- Expert Systems
- IoT-enabled Health Monitoring
- Edge AI for decision making

AI Technologies Used:

- Machine Learning: Learns daily routines and predicts health risks.
- Natural Language Processing (NLP): Enables voice communication between the robot and the elderly.
- Expert System: Uses medical knowledge rules to generate healthcare recommendations.
- IoT : Monitor heart rate, body temperature, oxygen level, and movement.
- Cloud Computing: Stores health records for long-term analysis.

Working Mechanism:

1. Health sensors collect vital signs.

2. Data is processed and cleaned.
3. ML model helps in health conditions.
4. Expert System applies medical rules.
5. Robot reminds users to take medicines.
6. NLP allows natural voice interaction.
7. In emergencies (fall or abnormal vitals), caregivers receive instant alerts.
8. Health data is securely stored in the cloud.

Literature Review:

Recent research highlights:

- AI improves medication adherence and patient monitoring.
- IoT-based system helps enable continuous health tracking.
- NLP enhances elderly–robot interaction.
- Expert Systems provide explainable medical recommendations.
- Current systems lack personalized decision-making, which this proposed system addresses.

Real-World Implementation, Impact & Challenges:

Example:

An elderly patient living alone forgets morning medication. The robot provides a voice reminder. If no response is detected and health sensors detect an abnormal heart rate or a fall, the system automatically alerts family members.

Applications:

- Smart Homes
- Hospitals
- Assisted Living Centers
- Rehabilitation Centers

Advantages:

- Personalized medication reminders
- Continuous health monitoring
- Early emergency detection
- Reduced caregiver workload
- Improves safety
- Supports independent living

Challenges

Future Scope:

- Integration with wearable devices
- Emotion recognition using AI
- Predictive disease diagnosis
- Generative AI for natural conversations
- Smart home automation integration

Originality:

This robot:-

- Learns user habits using Machine Learning.
- Uses NLP for natural communication.
- Combines Expert Systems with IoT for explainable healthcare decisions.
- Provides real-time emergency detection and healthcare assistance.

Conclusion:

The AI-Based Elderly Companion Robot is a modern intelligent healthcare solution that combines AI, ML, NLP, Expert Systems, and IoT to improve medication adherence, monitor health conditions, detect emergencies, and support independent living. The proposed system ensures patient safety, reduces caregiver burden, and represents a future AI-driven elderly healthcare.